

Practice — Keeping Time: Day, Month, Year and Calendars

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ**[1]**

The day is based on the Earth's —

- (a) revolution around the Sun (b) rotation on its axis (c) tilt (d) Moon's orbit

Q2 · MCQ**[1]**

A month is based on the —

- (a) Sun's rotation (b) Moon's cycle of phases (c) seasons (d) tides

Q3 · MCQ**[1]**

A solar year is about —

- (a) 29.5 days (b) 100 days (c) $365\frac{1}{4}$ days (d) 24 hours

Q4 · MCQ**[1]**

A stick's shadow is shortest —

- (a) at sunrise (b) at midday (Sun highest) (c) at sunset (d) at night

Q5 · ASSERTION-REASON**[1]**

A: A leap year has an extra day.

R: A year is about $365\frac{1}{4}$ days, so the quarter-days add up to one day every four years.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

A calendar based on the Moon's phases is a _____ calendar.

Q7 · FILL IN THE BLANK

[1]

A device that tells time using a shadow is a _____.

Short answer (2 marks each)

Q8

[2]

Name the natural cycle behind the day, the month and the year.

Q9

[2]

How can the length of a stick's shadow be used to find midday?

Q10

[2]

Differentiate between a lunar and a solar calendar.

Long / application (3 marks each)

Q11

[3]

Explain how the periodic motions in the sky give us the units day, month and year, with their approximate durations.

Q12

[3]

Why do calendars need adjustments such as leap years and lunisolar systems?

[Check yourself](#) → Answer key